DICOM Conformance Statement

MacAngioView — DICOM cine viewer Version 1.3b · Document date 2026-08-07 · Document version 2.7 (draft)

> **Disclaimer.** MacAngioView is a beta application for evaluation, review, and educational use only. It is **not a certified medical device** and is not intended for primary diagnosis or clinical decision-making. This document is structured after DICOM PS3.2 but has not been independently audited.

---

1. Conformance Statement Overview

MacAngioView is a standalone desktop application that reads DICOM PS3.10 (Part 10) files and DICOMDIR file-sets from local storage or mounted removable media (CD/DVD/USB), and displays X-ray angiographic (XA) cine runs, multiframe/single-frame ultrasound (US) images, and the Encapsulated PDF and Structured Report (SR) documents contained in a study. It can export loaded runs to QuickTime (.mov) movies and copy the displayed frame to the system clipboard (both non-DICOM outputs).

MacAngioView implements **Media Interchange services only**, in the **File-Set Reader (FSR)** role. It provides **no network services** of any kind.

Table 1-1. Network Services

Service	User of Service (SCU)	Provider of Service (SCP)
All DIMSE services (Verification, Storage, Query/Retrieve, Print, Storage Commitment, Worklist, MPPS)	No	No
DICOMweb (STOW-RS / QIDO-RS / WADO-RS)	No	No

The application never opens a network association or socket for DICOM purposes; it runs fully offline.

Table 1-2. Media Services

Media Storage Application Profile	Write Files (FSC/FSU)	Read Files (FSR)
General Purpose CD-R / DVD (STD-GEN-CD, STD-GEN-DVD-JPEG)	No	Yes
Basic Cardiac X-Ray Angiographic (STD-XABC-CD)	No	Yes
1024 X-Ray Angiographic (STD-XA1K-CD, STD-XA1K-DVD)	No	Yes
Ultrasound (STD-US-ID/SC/CC-MF…, CD and DVD media)	No	Yes

MacAngioView never creates or updates a DICOM file-set and never writes DICOM objects.

---

2. Introduction

2.1 Revision History

Doc version	Date	Application version	Notes
1.0 (draft)	2026-07-15	0.2.1b	Initial statement
1.1 (draft)	2026-07-15	0.2.2b	Frame VOI LUT functional group, Pixel Aspect Ratio, lossy compression indication
1.2 (draft)	2026-07-15	0.2.2b	Video transfer syntaxes (MPEG2/H.264/HEVC) decoded via bundled ffmpeg
1.3 (draft)	2026-07-16	0.3.0b	Structured Report documents rendered as paginated text
1.4 (draft)	2026-07-18	0.3.0b	Modality LUT Sequence, VOI LUT Sequence, VOI LUT Function (LINEAR_EXACT/SIGMOID)
1.5 (draft)	2026-07-18	0.3.0b	Positioner angles displayed; Person Name ideographic/phonetic groups formatted
1.6 (draft)	2026-07-18	0.3.0b	Frame Time Vector variable timing reproduced in playback
1.7 (draft)	2026-07-18	0.3.0b	Overlay Planes (60xx) and Display Shutter rendered
1.8 (draft)	2026-07-18	0.3.0b	HTJ2K transfer syntaxes decoded (pylibjpeg-openjpeg)
1.9 (draft)	2026-07-18	0.3.0b	XA mask subtraction (DSA, AVG_SUB) with View menu toggle
2.0 (draft)	2026-07-18	0.3.0b	User-designated DSA mask frame (application feature)
2.1 (draft)	2026-07-18	0.3.0b	Mask Sub-pixel Shift applied; user mask re-registration (nudge)
2.2 (draft)	2026-07-18	0.4.0b	Release 0.4.0b — revisions 1.4–2.1 ship in this application version
2.3 (draft)	2026-07-19	1.0b	US studies displayed at native matrix (rectangular viewport); zoom and thumbnails preserve source aspect; SR zero-valued NUM measurements rendered
2.4 (draft)	2026-07-23	1.1b	Documentation only — product description made platform-neutral ahead of the Windows build; clipboard described as the system clipboard. No change to supported SOP classes, transfer syntaxes, or the display pipeline
2.5 (draft)	2026-07-29	1.2b	Documentation only — Windows export folder pickers fixed and batch export now isolates per-run failures. No change to supported SOP classes, transfer syntaxes, or the display pipeline
2.6 (draft)	2026-08-04	1.2b	Documentation only — corrected the JPEG Extended (Process 2 & 4) transfer syntax claim: 12-bit samples are not currently decodable by either bundled decoder (python-

Doc version	Date	Application version	Notes
			gdcm, Pillow), verified against a real 12-bit XA sample. No change to the application itself
2.7 (draft)	2026-08-07	1.3b	Caliper Tool: application-computed distance measurements from Ultrasound Region / Pixel Spacing / Imager Pixel Spacing attributes, or a manual catheter calibration (§4.6, §7.1) — supersedes the 2.2–2.6 "calibrated measurements: not applied" note. Quiz Mode (§4.7): a non-DICOM self-study workflow, no change to the data read or displayed
2.8 (draft)	2026-08-10	1.4b	X-Ray Radiofluoroscopic Image Storage (RF) named explicitly as a supported SOP Class — decodes and displays through the same pipeline as XA; no application change was required (§4.4)
2.9 (draft)	2026-08-10	1.4b	Waveform Storage instances (12-lead/general/ambulatory ECG, hemodynamic, cardiac EP, ...) now rendered as paginated, calibrated lead-strip images (waveform_render) instead of only being listed by SOP Instance UID — supersedes the 2.2–2.8 "waveforms: not interpreted as such" note (§4.4)
2.10 (draft)	2026-08-23	1.4b	CT Image Storage (CT/CTA) named explicitly as a supported SOP Class: a study's single-frame CT slices sharing one Series Instance UID are grouped into one browsable multi-slice run (stepped, not cine-played) instead of arriving as N separate unplayable single-frame runs; multi-valued Window Center/Width (e.g. SOFT TISSUE/BONE/LUNG) is now offered as a selectable preset for any grayscale run that carries more than one W/L pair, not just CT (§4.4, §4.6, §7.1)

2.2 Audience and Remarks

This document is intended for people integrating or evaluating MacAngioView alongside other DICOM applications (modalities, media creators, PACS export functions). It assumes familiarity with the DICOM standard (PS3.1–PS3.20).

Conformance to this statement does not guarantee interoperability; validation with representative media from the intended source systems is recommended.

2.3 Definitions and Abbreviations

FSR — File-Set Reader. TS — Transfer Syntax. SOP — Service-Object Pair. VOI — Value of Interest. XA — X-Ray Angiography. US — Ultrasound.

2.4 Implementation Identification

Application name	MacAngioView
Version	1.3b
Platform	macOS 12+ (Intel and Apple Silicon via Rosetta 2); Windows 10/11 (64-bit)
DICOM toolkit	pydicom 3.0.2 with python-gdcm 3.2.6, pylibjpeg 2.1 (openjpeg plugin), and Pillow pixel-data decoders

3. Networking

Not applicable. MacAngioView implements no DICOM network application entities, initiates no associations, and accepts none.

---

4. Media Interchange

4.1 Implementation Model

One local Application Entity, **MEDIA-FSR**, is activated by the user through the GUI (File ▸ Open File…/Open Folder…, or the equivalent buttons). It reads a file-set or individual files, builds an index of the image and document instances found (metadata only), and then decodes the pixel data of a run on demand when the user selects it. Decoded frames are held in memory only; nothing is written back to the media, which can be ejected from within the application.

4.2 Real-World Activities

- **RWA 1 — Load a file-set:** the user selects a disc/folder. The application locates a `DICOMDIR` at the selected location or one subfolder level below it (case-insensitive name match) and enumerates every existing file referenced by Referenced File ID (0004,1500). Directory-record hierarchy attributes are not otherwise interpreted.
- **RWA 2 — Load individual files:** the user selects a single Part 10 file, or a folder containing `*.dcm` files (searched recursively) without a DICOMDIR.
- **RWA 3 — Review:** indexed instances appear as thumbnails; the selected instance is displayed with cine playback, frame stepping, window/level, zoom, and edge-enhancement controls. Encapsulated PDF instances are rendered as pageable images (or opened in an external PDF viewer), and Structured Report instances as pageable text renderings of their content tree.

4.3 Media Application Profiles — Roles

MacAngioView is a **general-purpose FSR**: it does not verify that a file-set declares or conforms to a specific application profile. Any file-set whose instances use the SOP classes and transfer syntaxes listed below can be read; this covers the STD-GEN, STD-XABC, STD-XA1K, and STD-US family profiles listed in Table 1-2.

4.4 SOP Classes

Instance selection is **content-based, not SOP-Class-restricted**: any instance whose pixel data can be decoded is displayed. The SOP classes below are the design targets and the ones exercised in testing.

SOP Class	UID	Support
X-Ray Angiographic Image Storage	1.2.840.10008.5.1.4.1.1.12.1	Display, cine
Enhanced XA Image Storage	1.2.840.10008.5.1.4.1.1.12.1.1	Display, cine (frame timing from per-frame Frame Content macros)

SOP Class	UID	Support
X-Ray Radiofluoroscopic Image Storage (RF)	1.2.840.10008.5.1.4.1.1.12.2	Display, cine — same pipeline as XA (content-based selection; not a distinct code path)
CT Image Storage	1.2.840.10008.5.1.4.1.1.2	Single-frame slices sharing a Series Instance UID are grouped into one browsable multi-slice run, ordered by Instance Number, and stepped (Left/Right or the frame slider) — not cine-played, since the "frames" are separate files with no playback timing of their own (§4.6)
Ultrasound Image Storage	1.2.840.10008.5.1.4.1.1.6.1	Display (single frame)
Ultrasound Multi-frame Image Storage	1.2.840.10008.5.1.4.1.1.3.1	Display, cine
Encapsulated PDF Storage	1.2.840.10008.5.1.4.1.1.104.1	Rendered per page (also detected by an `application/pdf` Encapsulated Document when the SOP Class UID is absent)
SR Document Storage family (Basic Text, Enhanced, Comprehensive, …)	1.2.840.10008.5.1.4.1.1.88.*	Content tree rendered as read-only paginated text (also detected by Modality SR with a Content Sequence when the SOP Class UID is absent). TEXT/CODE/NUM/PNAME/DATE-TIME/UIDREF items shown; IMAGE/COMPOSITE/WAVEFORM references listed by SOP Instance UID, not resolved
Waveform Storage family (12-lead/General/Ambulatory ECG, Hemodynamic, Cardiac Electrophysiology, …)	1.2.840.10008.5.1.4.1.1.9.*	Each multiplex group's channels rendered as paginated, calibrated lead-strip images (waveform_render) — a channel's trace is in physical units (e.g. µV) when Channel Sensitivity is present, else raw digital values; also detected by the presence of a Waveform Sequence when the SOP Class UID is absent
Secondary Capture and other image SOP classes	various	Best-effort display when pixel data is decodable
Media Storage Directory Storage (DICOMDIR)	1.2.840.10008.1.3.10	Read (Referenced File ID resolution only)

Presentation states (GSPS) and key object selections are **not** interpreted as such — but any image instance whose transfer syntax is a video codec (including US Multi-frame studies exported as MPEG-4) is decoded and played per 4.5. Structured Report documents are rendered as read-only paginated text of their content tree; coded entries are shown by their Code Meaning (coding scheme designators and values are not displayed), and templates are not validated. IMAGE/COMPOSITE references inside an SR content tree are listed by SOP Instance UID rather than resolved to the referenced instance, even when that instance is a Waveform run present elsewhere in the same study.

4.5 Transfer Syntaxes

Decoding is provided by pydicom with the python-gdcm and Pillow backends; the video transfer syntaxes are decoded by the bundled ffmpeg (imageio-ffmpeg). All decoders ship in the distributed application.

Transfer Syntax	UID	Read
Implicit VR Little Endian	1.2.840.10008.1.2	Yes
Explicit VR Little Endian	1.2.840.10008.1.2.1	Yes
Deflated Explicit VR Little Endian	1.2.840.10008.1.2.1.99	Yes
Explicit VR Big Endian (retired)	1.2.840.10008.1.2.2	Yes
RLE Lossless	1.2.840.10008.1.2.5	Yes
JPEG Baseline (Process 1)	1.2.840.10008.1.2.4.50	Yes
JPEG Extended (Process 2 & 4)	1.2.840.10008.1.2.4.51	8-bit only — both bundled decoders (python-gdcm, Pillow) refuse 12-bit samples; a 12-bit XA sample also failed against the (unbundled) pylibjpeg-libjpeg plugin
JPEG Lossless (Process 14)	1.2.840.10008.1.2.4.57	Yes
JPEG Lossless SV1 (Process 14, FOP=1)	1.2.840.10008.1.2.4.70	Yes
JPEG-LS Lossless / Near-Lossless	1.2.840.10008.1.2.4.80 / .81	Yes
JPEG 2000 Lossless / Lossy	1.2.840.10008.1.2.4.90 / .91	Yes
MPEG2, MPEG-4 AVC/H.264 (all profiles)	1.2.840.10008.1.2.4.100–.106	Yes — bundled ffmpeg
HEVC/H.265	1.2.840.10008.1.2.4.107 / .108	Yes — bundled ffmpeg
High-Throughput JPEG 2000 (HTJ2K)	1.2.840.10008.1.2.4.201–.203	Yes — pylibjpeg-openjpeg

Video-transfer-syntax runs are decoded whole into memory as RGB frames (the encapsulated bitstream is reconstructed across fragments), capped at 1000 frames per run; window/level does not apply to them. When the data set carries no frame-rate attribute, the rate reported by the video stream itself is used. An instance in an unsupported transfer syntax is indexed but its thumbnail shows **Error** and the failure is logged to the in-app Console; it does not block the rest of the study.

4.6 Display Pipeline (Attribute Usage)

- **Photometric Interpretation:** MONOCHROME1 (rendered inverted), MONOCHROME2, PALETTE COLOR (color LUT applied), RGB, and the YBR variants produced by the JPEG codecs (converted to RGB by the decoder).

- **Modality LUT:** a Modality LUT Sequence (0028,3000) is applied to grayscale data when present; otherwise the linear Rescale Slope (0028,1053) / Intercept (0028,1052).
- **VOI:** grayscale images are windowed with the VOI LUT Function (0028,1056) the data set declares — LINEAR, LINEAR_EXACT, or SIGMOID (PS3.3 C.11.2). The default window is resolved from, in order: the first value pair of the top-level Window Center (0028,1050) / Window Width (0028,1051); the Frame VOI LUT functional-group macro (0028,9132) — shared group first, then the first per-frame item (per-frame *varying* windows are not reproduced); the input range of the first VOI LUT Sequence (0028,3010) item; the full data range. When a VOI LUT Sequence is present its first LUT is rendered as the default view (output normalized over the table's value range); adjusting window/level interactively replaces the table with a window, and Reset returns to the table.
- **Pixel geometry:** non-square pixels are displayed square. The vertical:horizontal ratio is taken from Pixel Aspect Ratio (0028,0034), or from the row:column ratio of Pixel Spacing (0028,0030) / Imager Pixel Spacing (0018,1164) / the Enhanced shared Pixel Measures group; the lower-resolution axis is upscaled (never downsampled). Only the ratio is used here — see Calibrated distance measurement below for the one place an absolute spacing value is used.
- **Lossy compression indication:** the Lossy Image Compression state (0028,2110), with method (0028,2114) and ratio (0028,2112) when present, is shown in the Patient panel for the selected instance.
- **Cine timing:** the playback rate is resolved from, in order: XA Acquisition Frame Rate → Cine Rate → Recommended Display Frame Rate → Frame Time → mean of Frame Time Vector → Actual Frame Duration → for Enhanced multiframe objects, the median inter-frame interval of the per-frame Frame Acquisition/Reference DateTime values. A Frame Time Vector's per-frame variable timing is reproduced during playback: each frame's dwell is its vector entry, scaled proportionally when the user adjusts the rate (whose readout stays the mean rate). Non-positive entries fall back to the mean interval. QuickTime export renders at a single uniform rate.
- **Overlay Planes:** the 60xx Overlay Plane groups (Overlay Data bitmaps) are composited in white onto monochrome images after the VOI stage. A single-frame overlay applies to every frame of a multi-frame image; a multi-frame overlay maps through Image Frame Origin; Overlay Origin places the plane (clipped at the image edges). Overlays embedded in unused Pixel Data bits (retired in DICOM 2004) are not extracted, and overlays are not composited onto color images.
- **Display Shutter:** RECTANGULAR, CIRCULAR, and POLYGONAL shutter shapes (PS3.3 C.7.6.11) are combined and the area outside them is painted with the Shutter Presentation Value (high byte; black when absent) on monochrome images. The Enhanced multi-frame Frame Display Shutter macro is not applied.
- **Positioner angles:** Positioner Primary/Secondary Angle (0018,1510/ 0018,1511) — top-level or the Enhanced XA/XRF Positioner macro (0018,9405) — are shown in the Patient panel in the LAO/RAO · CRA/CAU convention (PS3.3 C.8.7.5). A rotational acquisition's per-frame varying angles are not tracked; the first value stands for the run.
- **Mask subtraction (DSA):** a Mask Subtraction Sequence (0028,6100) item with Mask Operation AVG_SUB enables the View ▸ Subtracted (DSA) toggle: the mask is the mean of the Mask Frame Numbers, subtraction is limited to the Applicable Frame Range (other frames show native), and a run whose Recommended Viewing Mode (0028,1090) is SUB starts subtracted. While subtracted, the window defaults span the subtracted signal; QuickTime export and Copy follow the on-screen state. Mask Sub-pixel Shift (0028,6114) is applied as the mask's initial registration shift (bilinear, edge-

replicated), and the user can re-register the mask against patient motion in 1 px steps (⌥-arrows, View ▸ Nudge DSA Mask) or 0.1 px steps (⇧⌥-arrows). Mask Operation TID is **not** applied. As an application feature beyond the data set, the user may designate the currently displayed frame as the mask (View menu), which enables subtraction for multi-frame grayscale cines carrying no Mask Subtraction Sequence and supersedes a file-provided spec until removed.

- **Multi-slice series grouping (CT/CTA):** single-frame image instances that share a Series Instance UID (0020,000E) — a CT/CTA axial series, which arrives as one file per slice with no cine mechanism of its own — are collapsed at index time into one browsable multi-slice run, ordered by Instance Number (0020,0013), falling back to SOP Instance UID then the source path on ties or when it is absent. Each slice is decoded from its own file on selection (no combined in-memory volume, no interpolation between slices — the same "decode on demand" pattern used for cine frames). The run steps (Left/Right, the frame slider) but does not cine-play, since there is no real inter-slice playback timing; it keeps window leveling and behaves like a normal image run for Quiz Mode, export, and the Caliper Tool — a true multi-frame cine (Number of Frames > 1) is never grouped this way, even when it shares a Series Instance UID with other runs.
- **Selectable Window Center/Width presets:** any grayscale run whose Window Center (0028,1050) / Window Width (0028,1051) carry more than one value pair — e.g. CT's SOFT TISSUE/BONE/LUNG presets — offers a Preset picker (info column) that applies a chosen pair as the active window; labels come positionally from Window Center & Width Explanation (0028,1055) when present, else "Preset N". The default window shown on first display is still the first pair, per the VOI resolution order above; selecting a preset does not persist across slices of a grouped series run (§ above) — stepping to another slice keeps the last window/level the user set, matching how a real DICOM viewer's W/L control behaves when paging a stack.
- **Calibrated distance measurement (Caliper Tool):** an application feature (View ▸ Caliper Tool) that reports a user-drawn two-point distance in millimeters, resolved per run at index time in priority order: (1) the first item of the Sequence of Ultrasound Regions (0018,6011)'s Physical Delta X (0018,602C) when its Physical Units X Direction (0018,6024) is cm; (2) Pixel Spacing (0028,0030)'s column value; (3) for XA/RF only, Imager Pixel Spacing (0018,1164) divided by the geometric magnification — Estimated Radiographic Magnification Factor (0018,1114), else Distance Source to Detector (0018,1110) ÷ Distance Source to Patient (0018,1111) — an isocenter-plane approximation shown as an **approximate** reading ("~N mm") until the user calibrates on a visible reference (below). No calibration is written back to the data set, and no DICOM measurement/annotation object (e.g. an SR or GSPS) is created — the result is display-only, shown as an on-screen overlay and optionally recorded in a Quiz Mode finding (§4.7). As an application feature beyond the data set, View ▸ Calibrate Caliper… lets the user re-derive the scale by measuring across a reference of known size (a catheter shaft, by French size, or a custom length) — this manual calibration always takes priority over the automatic scale above, persists per run for the session, and is the only path to a scale for a run whose data set has none of the attributes listed above.
- **Not applied / not displayed:** Grayscale Softcopy Presentation States.
- **Rendering:** frames are rendered to 8-bit for display, fit to a 512×512 or 1024×1024 viewport (aspect preserved, never magnified past 1:1 in 1024 mode). A study whose image instances are all Ultrasound is instead displayed in a viewport matching the study's largest pixel-aspect-corrected matrix (height rounded up to a multiple of 4; minimum 480×432, maximum 1024 per axis), so US frames render 1:1;

thumbnails and interactive zoom selections preserve the source aspect ratio. No PS3.14 (GSDF) display calibration is performed.

4.7 Extensions Beyond Part 10 Reading

- Files lacking File Meta Information (raw data sets) are accepted: the transfer syntax is inferred from the detected byte order / VR encoding and the data set is read best-effort.
- `DICOMDIR` located one directory level below the selected folder is accepted (some media place the file-set in a subfolder).
- Encapsulated PDF documents are rendered in-app via PyMuPDF.
- Loaded runs can be exported to QuickTime (.mov, H.264) via the bundled ffmpeg; the displayed frame can be copied to the clipboard. These outputs are **not** DICOM objects and carry no DICOM header; any burned-in identification in the pixel data is preserved as-is.
- **Quiz Mode** (View ▸ Quiz Mode) is a non-DICOM, application-only self-study feature: it withholds the display of a study's own SR/Encapsulated PDF instances from the Select strip until the user reveals them, and lets the user record structured findings (Cath/Echo/ Vascular templates) referencing the Caliper Tool's measurements. Neither toggling it nor recording findings reads any additional DICOM attribute or alters the source data set; the saved findings are written as a plain JSON file (not a DICOM SR or other DICOM object) alongside a de-identified run label and the application version, at a location the user chooses.

---

5. Support of Character Sets

Data-set text is decoded by pydicom according to Specific Character Set (0008,0005), including ISO-IR 100 (Latin-1) and the other single-byte and multi-byte character sets pydicom supports; decoded text is displayed as Unicode. Person Name values are displayed with component delimiters (`^`) replaced by spaces (empty components dropped); the ideographic and phonetic component groups (`=` delimited, PS3.5 6.2.1) follow the first group in parentheses, e.g. "Yamada Tarou (, やまだ たろう)".

---

6. Security

- The application runs entirely offline; no data leaves the machine except by explicit user action (QuickTime export, clipboard copy).
- Media are read-only to the application; Eject unmounts the volume and purges all decoded frames and demographics from memory.
- Decoded pixel data is held in memory only; no image cache is written to disk. Two transient exceptions: an Encapsulated PDF opened externally is written as a temporary PDF file for the system viewer, and a video-transfer-syntax bitstream is written to a temporary file for the ffmpeg decoder and deleted as soon as decoding ends.
- The application is Developer ID signed and Apple-notarized. The beta build stops launching 90 days after first launch.

- No de-identification is performed on export/copy outputs.

---

7. Annexes

7.1 IOD Attribute Usage (informative)

Attributes read for indexing and display include: SOP Class UID, Transfer Syntax UID, Modality, Number of Frames, Rows/Columns, Bits Allocated/Stored, Photometric Interpretation, Pixel Data, the frame-rate attributes listed in 4.6, Window Center/Width, the Frame VOI LUT functional-group macro, Modality LUT Sequence, Rescale Slope/Intercept, VOI LUT Sequence, VOI LUT Function, Pixel Aspect Ratio, Pixel Spacing / Imager Pixel Spacing / Pixel Measures (ratio only, except where noted below), Lossy Image Compression / Method / Ratio, Palette Color LUT attributes, Positioner Primary/Secondary Angle (and the Positioner Position macro), the 60xx Overlay Plane attributes, the Display Shutter attributes, the Mask Subtraction Sequence and Recommended Viewing Mode, Patient Name, Patient ID, Patient Birth Date, Study Date, Study Time, Encapsulated Document, MIME Type of Encapsulated Document, Waveform Sequence and its Number of Waveform Channels/Samples, Sampling Frequency, Waveform Bits Allocated, Waveform Sample Interpretation, Waveform Data, Multiplex Group Label, and the per-channel Channel Definition Sequence (Channel Source Sequence, Channel Sensitivity, Channel Sensitivity Correction Factor, Channel Baseline, Channel Sensitivity Units Sequence, Channel Label), Series Instance UID, Instance Number and Window Center & Width Explanation (for multi-slice series grouping and W/L presets — see §4.6), and (DICOMDIR) Referenced File ID.

For the Caliper Tool's calibrated distance measurement (§4.6), Pixel Spacing and Imager Pixel Spacing are additionally read for their absolute value (not just the ratio), together with Sequence of Ultrasound Regions / Physical Delta X / Physical Units X Direction, Estimated Radiographic Magnification Factor, and Distance Source to Detector / Patient.

7.2 Grayscale Rendering (informative)

Grayscale frames are mapped with the DICOM VOI equations (PS3.3 C.11.2.1.2 LINEAR; C.11.2.1.3 LINEAR_EXACT and SIGMOID) over the effective window, or through the data set's VOI LUT table at the default window, then quantized to 8 bits; MONOCHROME1 output polarity is inverted after the VOI stage.